

PRAKTIKUM 1
STRUKTUR SISTEM OPERASI



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1. OPERATING-SYSTEM SERVICES

1.1. Menjelajahi Layanan OS

- Lihat informasi sistem

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ uname -a
Linux breaksek 6.12.13-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.13-1kali1
NU/Linux

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lsb_release -a
No LSB modules are available.
Distributor ID: Kali
Description:    Kali GNU/Linux Rolling
Release:        2025.3
Codename:       kali-rolling
```

- Lihat proses yang berjalan

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ps aux | head -20
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.2	25020	15188	?	Ss	18:11	0:02	/sbin/init splash
root	2	0.0	0.0	0	0	?	S	18:11	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	S	18:11	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	18:11	0:00	[kworker/R-rcu_gp]
root	5	0.0	0.0	0	0	?	I<	18:11	0:00	[kworker/R-sync_wq]
root	6	0.0	0.0	0	0	?	I<	18:11	0:00	[kworker/R-slub_flushwq]
root	7	0.0	0.0	0	0	?	I<	18:11	0:00	[kworker/R-netns]
root	9	0.0	0.0	0	0	?	I<	18:11	0:00	[kworker/0:0H-events_highpri]
root	10	0.0	0.0	0	0	?	I	18:11	0:02	[kworker/0:1-pm]
root	12	0.0	0.0	0	0	?	I<	18:11	0:00	[kworker/R-mm_percpu_wq]
root	13	0.0	0.0	0	0	?	I	18:11	0:00	[rcu_tasks_kthread]
root	14	0.0	0.0	0	0	?	I	18:11	0:00	[rcu_tasks_rude_kthread]
root	15	0.0	0.0	0	0	?	I	18:11	0:00	[rcu_tasks_trace_kthread]
root	16	0.0	0.0	0	0	?	S	18:11	0:00	[ksoftirqd/0]
root	17	0.1	0.0	0	0	?	I	18:11	0:05	[rcu_preempt]
root	18	0.0	0.0	0	0	?	S	18:11	0:00	[rcu_exp_par_gp_kthread_worker/0]
root	19	0.0	0.0	0	0	?	S	18:11	0:00	[rcu_exp_gp_kthread_worker]
root	20	0.0	0.0	0	0	?	S	18:11	0:00	[migration/0]
root	21	0.0	0.0	0	0	?	S	18:11	0:00	[idle_inject/0]

- Lihat penggunaan memori

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	5.7Gi	2.5Gi	265Mi	240Mi	3.4Gi	3.2Gi
Swap:	5.9Gi	0B	5.9Gi			

- Lihat informasi CPU

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lscpu
```

Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 36 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 4
On-line CPU(s) list: 0-3
Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM) i7-3667U CPU @ 2.00GHz
CPU family: 6
Model: 58
Thread(s) per core: 2
Core(s) per socket: 2
Socket(s): 1
Stepping: 9
CPU(s) scaling MHz: 26%
CPU max MHz: 3200.0000
CPU min MHz: 800.0000
BogoMIPS: 4988.26
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat p
se36 clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx rdt
scp lm constant_tsc arch_perfmon pebs bts rep_good nopl xtopology n
onstop_tsc cpuid aperfmperf pni pclmulqdq dtes64 monitor ds_cpl vmx
smx est tm2 ssse3 cx16 xtpr pdcm pcid sse4_1 sse4_2 x2apic popcnt
tsc_deadline_timer aes xsave avx f16c rdrand lahf_lm cpuid_fault ep
b pti ssbd ibrs ibpb stibp tpr_shadow flexpriority ept vpid fsgsbas
e smep erms xsaveopt dtherm ida arat pln pts vnmi md_clear flush_l1
d

Virtualization features:
Virtualization: VT-x
Caches (sum of all):
L1d: 64 KiB (2 instances)
L1i: 64 KiB (2 instances)
L2: 512 KiB (2 instances)
L3: 4 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-3
Vulnerabilities:
Gather data sampling: Not affected
Itlb multihit: KVM: Mitigation: VMX disabled
L1tf: Mitigation; PTE Inversion; VMX conditional cache flushes, SMT vulne
rable
Mds: Mitigation; Clear CPU buffers; SMT vulnerable
Meltdown: Mitigation; PTI
Mmio stale data: Unknown: No mitigations
Reg file data sampling: Not affected
Retbleed: Not affected
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitizatio
n
Spectre v2: Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP conditiona
l; RSB filling; PBRSE-eIBRS Not affected; BHI Not affected
Srbds: Vulnerable: No microcode
Tsx async abort: Not affected

- Lihat perangkat yang terpasang

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0   55.5M 1 loop /snap/core18/2952
loop1       7:1      0   73.9M 1 loop /snap/core22/2111
loop2       7:2      0   55.5M 1 loop /snap/core18/2947
loop3       7:3      0   73.9M 1 loop /snap/core22/2133
loop4       7:4      0    4K 1 loop /snap/bare/5
loop5       7:5      0   66.8M 1 loop /snap/core24/1151
loop6       7:6      0   66.8M 1 loop /snap/core24/1196
loop7       7:7      0   67.2M 1 loop /snap/cups/1100
loop8       7:8      0  618.3M 1 loop /snap/gnome-46-2404/125
loop9       7:9      0  589.6M 1 loop /snap/gnome-46-2404/117
loop10      7:10     0   47.6M 1 loop /snap/cups/1112
loop11      7:11     0  164.8M 1 loop /snap/gnome-3-28-1804/198
loop12      7:12     0  290.8M 1 loop /snap/mesa-2404/887
loop13      7:13     0   91.7M 1 loop /snap/gtk-common-themes/1535
loop14      7:14     0  290.8M 1 loop /snap/mesa-2404/912
loop15      7:15     0  344.9M 1 loop /snap/metasploit-framework/2137
loop16      7:16     0  345.6M 1 loop /snap/metasploit-framework/2173
loop17      7:17     0   93.8M 1 loop /snap/scrcpy/457
loop18      7:18     0    7.2M 1 loop /snap/ngrok/315
loop19      7:19     0    7.2M 1 loop /snap/ngrok/310
loop20      7:20     0  844.3M 1 loop /snap/onlyoffice-desktopeditors/655
loop21      7:21     0  840.8M 1 loop /snap/onlyoffice-desktopeditors/604
loop22      7:22     0   93.8M 1 loop /snap/scrcpy/460
loop23      7:23     0   49.3M 1 loop /snap/snapd/24792
loop24      7:24     0   940K 1 loop /snap/speedtest/9
loop25      7:25     0   50.8M 1 loop /snap/snapd/25202
sda         8:0      0  119.2G 0 disk 
├─sda1      8:1      0   976M 0 part /boot/efi
├─sda2      8:2      0  112.4G 0 part /
└─sda3      8:3      0    5.9G 0 part [SWAP]
```

1.2. Program C - Menggunakan Layanan OS

- Membuat, mengcompile serta menjalankan scanning service operating sistem

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano pelayanan_os.c
[sudo] password for rif:

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo gcc -o pelayanan_os pelayanan_os.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ./pelayanan_os
=== Program Execution Service ===
Process ID (PID): 6366
Parent Process ID (PPID): 4467

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 664

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
```

2. USER AND OS INTERFACE

2.1. Program C - Menggunakan Layanan OS

- Navigasi dasar

```

/home/rif/Desktop/filetugas/sistem_operasi
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ls -la
total 36
drwxrwxr-x 2 rif rif 4096 Oct 8 20:12 .
drwxrwxr-x 3 rif rif 4096 Oct 8 19:00 ..
-rwxr-xr-x 1 root root 16512 Oct 8 20:12 pelayanan_os
-rw-r--r-- 1 root root 1315 Oct 8 19:08 pelayanan_os.c
-rw-rw-r-- 1 rif rif 23 Oct 8 20:12 test_file.txt

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cd /tmp

(venv)(rif@breaksek)-[/tmp]
$ mkdir make_testdir

(venv)(rif@breaksek)-[/tmp]
$ cd make_testdir/

(venv)(rif@breaksek)-[/tmp/make_testdir]
$

```

- Pipelining

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ls -l | grep "^d"

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ps aux | grep bash | wc -l
2

```

- Redirection

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ echo "Hello World" > output.txt

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat output.txt
Hello World

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat output.txt >> output.tx

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat < output.txt
Hello World

```

- Background processes

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sleep 30 &
[1] 6549

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ jobs
[1]+  Running                  sleep 30 &

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ fg %1
sleep 30
```

2.2. Program C - Shell Sederha

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano shell_custom.c
[sudo] password for rif:

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -o shell_custom shell_custom.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ./shell_custom
myshell> help
execvp failed: No such file or directory
myshell> exit

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

3. SYSTEM CALLS

3.1. Process Control System Calls

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano proses_syscall.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -o proses_syscall proses_syscall.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ./proses_syscall
== Process Control System Calls ==
Parent PID: 7849
Parent waiting for child (PID: 7850) ...
Child PID: 7850
Child's Parent PID: 7849
Child executing 'ls' command ...
total 80
-rw-rw-r-- 1 rif rif 12 Oct 8 20:28 output.tx
-rw-rw-r-- 1 rif rif 12 Oct 8 20:27 output.txt
-rwxr-xr-x 1 root root 16512 Oct 8 20:12 pelayanan_os
-rw-r--r-- 1 root root 1315 Oct 8 19:08 pelayanan_os.c
-rwxrwxr-x 1 rif rif 16368 Oct 8 21:19 proses_syscall
-rw-r--r-- 1 root root 1042 Oct 8 21:16 proses_syscall.c
-rwxrwxr-x 1 rif rif 16552 Oct 8 21:04 shell_custom
-rw-rw-r-- 1 rif rif 1240 Oct 8 21:00 shell_custom.c
-rw-rw-r-- 1 rif rif 23 Oct 8 20:12 test_file.txt
Child exited with status: 0

```

3.2. File Management System Calls

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano file_syscall.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -o file_syscall file_syscall.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ./file_syscall
== File Management System Calls ==
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!

```


3.3. Information Maintenance System C

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano info_syscall.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -o info_syscall info_syscall.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ./info_syscall
=== Information Maintenance System Calls ===

Process ID: 8084
Parent Process ID: 4467
User ID: 1000
Group ID: 1000
Current time: Wed Oct  8 21:31:52 2025
Clock ticks since boot: 1719157458
Sleeping for 2 seconds ...
Awake!
```

4. SYSTEM SERVICES

4.1. Menggunakan systemd

- Lihat semua service

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ systemctl list-units --type=service
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
apparmor.service	loaded	active	exited	Load AppArmor profiles
colord.service	loaded	active	running	Manage, Install and Generate Color Profiles
console-setup.service	loaded	active	exited	Set console font and keymap
cron.service	loaded	active	running	Regular background program processing daemon
dbus.service	loaded	active	running	D-Bus System Message Bus
getty@tty1.service	loaded	active	running	Getty on tty1
haveged.service	loaded	active	running	Entropy Daemon based on the HAVEGE algorithm
ifupdown-pre.service	loaded	active	exited	Helper to synchronize boot up for ifupdown
keyboard-setup.service	loaded	active	exited	Set the console keyboard layout
kmod-static-nodes.service	loaded	active	exited	Create List of Static Device Nodes
lightdm.service	loaded	active	running	Light Display Manager
mariaDB.service	loaded	active	running	MariaDB 11.8.3 database server
ModemManager.service	loaded	active	running	Modem Manager
networking.service	loaded	active	exited	Raise network interfaces
NetworkManager-wait-online.service	loaded	active	exited	Network Manager Wait Online
NetworkManager.service	loaded	active	running	Network Manager
pcscd.service	loaded	active	running	PC/SC Smart Card Daemon
plymouth-quit-wait.service	loaded	active	exited	Hold until boot process finishes up
plymouth-read-write.service	loaded	active	exited	Tell Plymouth To Write Out Runtime Data
plymouth-start.service	loaded	active	exited	Show Plymouth Boot Screen
polkit.service	loaded	active	running	Authorization Manager
rpc-statd-notify.service	loaded	active	exited	Notify NFS peers of a restart
rtkit-daemon.service	loaded	active	running	RealtimeKit Scheduling Policy Service
snap.cups.cups-browsed.service	loaded	active	running	Service for snap application cups.cups-browsed
snap.cups.cupsd.service	loaded	active	running	Service for snap application cups.cupsd
snappd.apparmor.service	loaded	active	exited	Load AppArmor profiles managed internally by snappd
snappd.service	loaded	active	running	Snap Daemon
systemd-backlight@backlight:intel_backlight.service	loaded	active	exited	Load/Save Screen Backlight Brightness of backlight:intel_backlight
systemd-binfmt.service	loaded	active	exited	Set Up Additional Binary Formats
systemd-fsck@dev-disk-by\x2duuid-CB56\x2d2C49.service	loaded	active	exited	File System Check on /dev/disk/by-uuid/CB56-2C49
systemd-journal-flush.service	loaded	active	exited	Flush Journal to Persistent Storage
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-modules-load.service	loaded	active	exited	Load Kernel Modules
systemd-random-seed.service	loaded	active	exited	Load/Save OS Random Seed
systemd-remount-fs.service	loaded	active	exited	Remount Root and Kernel File Systems
systemd-sysctl.service	loaded	active	exited	Apply Kernel Variables
systemd-timesyncd.service	loaded	active	running	Network Time Synchronization

lines 1-40... skipping...

- Status service

```
(venv) rif@breaksek: ~/Desktop/filetugas/sistem_operasi
$ systemctl status mariadb
● mariadb.service - MariaDB 11.8.3 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-10-09 01:11:52 WIB; 3h 30min left
   Invocation: b9c65465db864cf97155a257d161a85
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 1251 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
   Process: 1257 ExecStartPre=/bin/sh -c '[ ! -e /usr/bin/galera_recovery ] && VAR=ll VAR=/usr/bin/galera_recovery'; [ $? -eq 0 ] && echo _WSREP_START_POSITION=$VAR > /run/mysqld/wsrep>
   Process: 1400 ExecStartPost=/bin/m -f /run/mysqld/wsrep-start-position (code=exited, status=0/SUCCESS)
   Process: 1415 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
   Main PID: 1330 (mariadb)
     Status: "Taking your SQL requests now..."
       Tasks: 10 (Limit: 4469)
      Memory: 153.5M (peak: 159.4M)
         CPU: 1.198s
      CGroup: /system.slice/mariadb.service
              └─1330 /usr/sbin/mariadb

Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: Buffer pool(s) load completed at 251009 1:11:51
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 0 [Note] mariadb: Event Scheduler: Loaded 0 events
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 0 [Note] /usr/sbin/mariadb: ready for connections.
Oct 09 01:11:52 breaksek mariadb[1330]: Version: '11.8.3-MariaDB-1+b1 from Debian' socket: '/run/mysqld/mysqld.sock' port: 3306 -- Please help get to 10k stars at https://github.com/Mari>
Oct 09 01:11:52 breaksek /etc/mysql/debian-start[1425]: Upgrading MariaDB tables if necessary.
Oct 09 01:11:52 breaksek systemd[1]: Started mariadb.service - MariaDB 11.8.3 database server.
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 3 [Warning] Access denied for user 'root'@'localhost' (using password: NO)
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 4 [Warning] Access denied for user 'root'@'localhost' (using password: NO)
Oct 09 01:11:52 breaksek debian-start[1442]: ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: NO)

(venv) rif@breaksek: ~/Desktop/filetugas/sistem_operasi
$
```

- Lihat log service

```
(venv) rif@breaksek: ~/Desktop/filetugas/sistem_operasi
$ journalctl -u mariadb -n 20
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: File system buffers for log disabled (block size=512 bytes)
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: End of log at LSN=638311
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: Opened 3 undo tablespaces
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: 128 rollback segments in 3 undo tablespaces are active.
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: Setting file './ibtmp1' size to 12.000MiB. Physically writing the file full; Please wait ...
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: File './ibtmp1' size is now 12.000MiB.
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: log sequence number 638311; transaction id 1469
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] Plugin 'FEEDBACK' is disabled.
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] Plugin 'wsrep-provider' is disabled.
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] InnoDB: Buffer pool(s) load completed at 251009 1:11:51
Oct 09 01:11:51 breaksek mariadb[1330]: 2025-10-09 1:11:51 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 0 [Note] mariadb: Event Scheduler: Loaded 0 events
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 0 [Note] /usr/sbin/mariadb: ready for connections.
Oct 09 01:11:52 breaksek mariadb[1330]: Version: '11.8.3-MariaDB-1+b1 from Debian' socket: '/run/mysqld/mysqld.sock' port: 3306 -- Please help get to 10k stars at https://github.com/Mari>
Oct 09 01:11:52 breaksek /etc/mysql/debian-start[1425]: Upgrading MariaDB tables if necessary.
Oct 09 01:11:52 breaksek systemd[1]: Started mariadb.service - MariaDB 11.8.3 database server.
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 3 [Warning] Access denied for user 'root'@'localhost' (using password: NO)
Oct 09 01:11:52 breaksek mariadb[1330]: 2025-10-09 1:11:52 4 [Warning] Access denied for user 'root'@'localhost' (using password: NO)
Oct 09 01:11:52 breaksek debian-start[1442]: ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: NO)
lines 1-20/20 (END)
```


- Lihat service yang berjalan saat boot

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ systemctl list-unit-files --type=service --state=enabled
```

UNIT FILE	STATE	PRESET
accounts-daemon.service	enabled	enabled
apparmor.service	enabled	disabled
console-setup.service	enabled	enabled
cron.service	enabled	enabled
e2scrub_reap.service	enabled	disabled
getty@.service	enabled	enabled
haveged.service	enabled	enabled
keyboard-setup.service	enabled	enabled
lightdm.service	enabled	disabled
mariadb.service	enabled	disabled
ModemManager.service	enabled	enabled
networking.service	enabled	enabled
NetworkManager-dispatcher.service	enabled	disabled
NetworkManager-wait-online.service	enabled	disabled
NetworkManager.service	enabled	enabled
regenerate-ssh-host-keys.service	enabled	enabled
snap.cups.cups-browsed.service	enabled	disabled
snap.cups.cupsd.service	enabled	disabled
snapt.apparmor.service	enabled	disabled
snapt.service	enabled	disabled
systemd-pstore.service	enabled	enabled
systemd-timesyncd.service	enabled	enabled

```
22 unit files listed.
```

4.2. Membuat Custom System Service

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano custom_daemon.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -o custom_daemon custom_daemon.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano /etc/systemd/system/custom_daemon.service

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo systemctl daemon-reload

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo systemctl start custom_daemon

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo systemctl status custom_daemon
o custom_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/custom_daemon.service; enabled; preset: disabled)
   Active: inactive (dead) since Wed 2025-10-08 22:29:27 WIB; 2s ago
     Duration: 56ms
  Invocation: ce2367abfaa14441a75236c3f730a33d
    Process: 9720 ExecStart=/home/rif/Desktop/filetugas/sistem_operasi/custom_daemon (code=exited, status=0/SUCCESS)
   Main PID: 9720 (code=exited, status=0/SUCCESS)
    Mem peak: 1.8M
      CPU: 22ms

Oct 08 22:29:27 breaksek systemd[1]: Started custom_daemon.service - My Custom Daemon.
Oct 08 22:29:27 breaksek custom_daemon[9722]: Custom Daemon is Started
Oct 08 22:29:27 breaksek custom_daemon[9722]: Daemon is running...
Oct 08 22:29:27 breaksek custom_daemon[9722]: Custom Daemon is Stopped
Oct 08 22:29:27 breaksek systemd[1]: custom_daemon.service: Deactivated successfully.
```

5. LINKERS AND LOADERS

5.1. Proses Linking

- Preprocessing

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano math_ops.h

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano math_ops.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano main.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -E main.c -o main.i

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -E math_ops.c -o math_ops.i

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ █

```

- Compilation (assembly)

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -S main.c -o main.s

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -S math_ops.c -o math_ops.s

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ █

```

- Assembly (object files)

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -c main.c -o main.o

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -c math_ops.c -o math_ops.o

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ █

```

- Linking

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc main.o math_ops.o -o program

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ █

```

- Lihat symbols dalam object file

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ nm main.o
0000000000000000 T main
0000000000000000 U multiply
0000000000000000 U printf

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ nm math_ops.o
0000000000000000 T add
0000000000000014 T multiply

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ nm program
00000000000002144 r __abi_tag
0000000000000118e T add
00000000000004018 B __bss_start
00000000000004018 b completed.0
00000000000004008 w __cxa_finalize@GLIBC_2.2.5
00000000000004008 D __data_start
00000000000004008 W data_start
00000000000001080 t deregister_tm_clones
000000000000010f0 t __do_global_ctors_aux
00000000000003dd8 d __do_global_ctors_aux_fini_array_entry
00000000000004010 D dso_handle
00000000000003de0 d _DYNAMIC
00000000000004018 D _edata
00000000000004020 B _end
000000000000011b8 T _fini
00000000000001130 t frame_dummy
00000000000003dd0 d __frame_dummy_init_array_entry
00000000000002140 r _FRAME_END__
00000000000003fe8 d _GLOBAL_OFFSET_TABLE__
0000000000000201c r __GNU_EH_FRAME_HDR
00000000000001000 T _init
00000000000002000 R _IO_stdin_used
00000000000002000 w _ITM_deregisterTMCloneTable
00000000000002000 w _ITM_registerTMCloneTable
00000000000002000 U __libc_start_main@GLIBC_2.34
00000000000001139 T main
000000000000011a2 T multiply
000000000000010b0 t register_tm_clones
00000000000001050 T _start
00000000000004018 D _TMC_END__

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$

```

- Lihat dependencies

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ldd program
linux-vdso.so.1 (0x00007fc907bf1000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007fc9079dc000)
/lib64/ld-linux-x86-64.so.2 (0x00007fc907bf3000)

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$

```

5.2. Static vs Dynamic Linking

- Static linking

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc main.c math_ops.c -o program_static -static

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$

```

- Dynamic linking (default)


```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc main.c math_ops.c -o program_dynamic

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

- Bandingkan ukuran

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ls -lh program_static program_dynamic
-rwxrwxr-x 1 rif rif 16K Oct  9 11:26 program_dynamic
-rwxrwxr-x 1 rif rif 737K Oct  9 11:25 program_static

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

- Lihat dependencies

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ldd program_static
not a dynamic executable

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ldd program_dynamic
linux-vdso.so.1 (0x00007fa1b6642000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007fa1b642d000)
/lib64/ld-linux-x86-64.so.2 (0x00007fa1b6644000)
```

5.3. Membuat dan Menggunakan Shared Library

- Compile sebagai shared library

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -c -fPIC math_ops.c -o math_ops.o

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -shared -o libmathops.so math_ops.o

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

- Compile program yang menggunakan library

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc main.c -L. -lmathops -o program_shared

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

- Jalankan dengan LD_LIBRARY_PATH


```
(venv)(rif@ breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15

(venv)(rif@ breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

- Lihat informasi loader

```
(venv)(rif@ breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ readelf -d program_shared

Dynamic section at offset 0x2dd0 contains 27 entries:
   Tag               Type                             Name/Value
0x0000000000000001 (NEEDED)                         Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)                         Shared library: [libc.so.6]
0x000000000000000c (INIT)                          0x1000
0x000000000000000d (FINI)                          0x11b0
0x0000000000000019 (INIT_ARRAY)                       0x3dc0
0x000000000000001b (INIT_ARRAYSZ)                     8 (bytes)
0x000000000000001a (FINI_ARRAY)                       0x3dc8
0x000000000000001c (FINI_ARRAYSZ)                     8 (bytes)
0x000000006ffffef5 (GNU_HASH)                       0x3b0
0x0000000000000005 (STRTAB)                          0x4b0
0x0000000000000006 (SYMTAB)                          0x3d8
0x000000000000000a (STRSZ)                            170 (bytes)
0x000000000000000b (SYMENT)                          24 (bytes)
0x0000000000000015 (DEBUG)                          0x0
0x0000000000000003 (PLTGOT)                          0x3fe8
0x0000000000000002 (PLTRELSZ)                       72 (bytes)
0x0000000000000014 (PLTREL)                          RELA
0x0000000000000017 (JMPREL)                          0x660
0x0000000000000007 (RELA)                            0x5a0
0x0000000000000008 (RELASZ)                          192 (bytes)
0x0000000000000009 (RELAENT)                         24 (bytes)
0x000000006ffffffb (FLAGS_1)                        Flags: PIE
0x000000006ffffffe (VERNEED)                        0x570
0x000000006fffffff (VERNEEDNUM)                     1
0x000000006ffffff0 (VERSYM)                          0x55a
0x000000006ffffff9 (RELACOUNT)                       3
0x0000000000000000 (NULL)                          0x0

(venv)(rif@ breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

6. OS-SPECIFIC APPLICATIONS

6.1. System Call Differences

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help
GNU nano 8.6
#include <stdio.h>
#ifdef __linux__
#include <unistd.h>
#include <sys/utsname.h>

void print_os_info() {
    struct utsname info;
    uname(&info);
    printf("OS: Linux\n");
    printf("System: %s\n", info.sysname);
    printf("Release: %s\n", info.release);
    printf("Version: %s\n", info.version);
}

#elif _WIN32
#include <windows.h>

void print_os_info() {
    printf("OS: Windows\n");
    // Windows-specific code
}

#else
void print_os_info() {
    printf("Unknown OS\n");
}

#endif

int main() {
    printf("== OS-Specific Application ==\n");
    print_os_info();

    #ifdef __linux__
    printf("\nLinux-specific: Process ID = %d\n", getpid());
    #endif

    return 0;
}
```

6.2. Binary Format Analysis

- Lihat format executable

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=e6987ed74a80a9c1881e4ad368f05f413dc1602d, for GNU/Linux 3.2.0, not stripped
```

- Lihat header ELF

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:          ELF64
  Data:            2's complement, little endian
  Version:         1 (current)
  OS/ABI:          UNIX - System V
  ABI Version:     0
  Type:            DYN (Position-Independent Executable file)
  Machine:         Advanced Micro Devices X86-64
  Version:         0x1
  Entry point address: 0x1050
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14064 (bytes into file)
  Flags:           0x0
  Size of this header:   64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 14
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
```

- Lihat section headers

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
$ readelf -S program
There are 31 section headers, starting at offset 0x36f0:

Section Headers:
 [Nr] Name              Type             Address           Offset
     Size              EntSize          Flags              Link            Info
-----
 [ 0]                     NULL              0000000000000000 00000000
 [ 1] .note.gnu.pr[...] NOTE              0000000000000350 00000350
 [ 2] .note.gnu.bu[...] NOTE              0000000000000370 00000370
 [ 3] .interp              PROGBITS          0000000000000394 00000394
 [ 4] .gnu.hash             GNU_HASH           00000000000003b0 000003b0
 [ 5] .dynsym               DYNSTR             00000000000003d8 000003d8
 [ 6] .dynstr               STRTAB             0000000000000480 00000480
 [ 7] .gnu.version          VERSYM             0000000000000510 00000510
 [ 8] .gnu.version_r        VERNEED            0000000000000520 00000520
 [ 9] .rela.dyn             RELA               0000000000000550 00000550
[10] .rela.plt             RELA               0000000000000610 00000610
[11] .init                 PROGBITS           0000000000001000 00001000
[12] .plt                 PROGBITS           0000000000001020 00001020
[13] .plt.got              PROGBITS           0000000000001040 00001040
[14] .text                 PROGBITS           0000000000001050 00001050
[15] .fini                 PROGBITS           00000000000011b8 000011b8
[16] .rodata               PROGBITS           0000000000002000 00002000
[17] .eh_frame_hdr         PROGBITS           000000000000201c 0000201c
[18] .eh_frame             PROGBITS           0000000000002058 00002058
[19] .note.ABI-tag         NOTE               0000000000002144 00002144
[20] .init_array           INIT_ARRAY         0000000000003dd0 00002dd0
[21] .fini_array           FINI_ARRAY         0000000000003dd8 00002dd8
[22] .dynamic               DYNAMIC            0000000000003de0 00002de0
```

- Lihat program headers

```

rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek) ~/Desktop/filetugas/sistem_operasi
$ readelf -l program

Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1050
There are 14 program headers, starting at offset 64

Program Headers:
Type           Offset             VirtAddr           PhysAddr
-----
PHDR           0x0000000000000040 0x0000000000000040 0x0000000000000040
INTERP         0x0000000000000310 0x0000000000000310 R 0x8
LOAD           0x0000000000000394 0x0000000000000394 0x0000000000000394
LOAD           0x00000000000001c 0x00000000000001c R 0x1
[Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
LOAD           0x0000000000000000 0x0000000000000000 0x0000000000000000
LOAD           0x0000000000000628 0x0000000000000628 R 0x1000
LOAD           0x0000000000001000 0x0000000000001000 0x0000000000001000
LOAD           0x0000000000001c1 0x0000000000001c1 R E 0x1000
LOAD           0x0000000000002000 0x0000000000002000 0x0000000000002000
LOAD           0x000000000000164 0x000000000000164 R 0x1000
LOAD           0x0000000000002dd0 0x0000000000003dd0 0x0000000000003dd0
DYNAMIC        0x000000000000248 0x000000000000250 RW 0x1000
DYNAMIC        0x0000000000002de0 0x0000000000003de0 0x0000000000003de0
NOTE           0x0000000000001e0 0x0000000000001e0 RW 0x8
NOTE           0x000000000000350 0x000000000000350 0x000000000000350
NOTE           0x000000000000020 0x000000000000020 R 0x8
NOTE           0x0000000000000370 0x0000000000000370 0x0000000000000370
NOTE           0x000000000000024 0x000000000000024 R 0x4
NOTE           0x0000000000002144 0x0000000000002144 0x0000000000002144
GNU_PROPERTY   0x0000000000000350 0x0000000000000350 0x0000000000000350
GNU_EH_FRAME   0x0000000000000020 0x0000000000000020 R 0x8
GNU_STACK      0x0000000000000000 0x0000000000000000 0x0000000000000000
GNU_RELRO      0x0000000000000000 0x0000000000000000 RW 0x10
GNU_RELRO      0x0000000000002dd0 0x0000000000003dd0 0x0000000000003dd0
GNU_RELRO      0x000000000000230 0x000000000000230 R 0x1

Section to Segment mapping:
Segment Sections ...
00
01 .interp
02 .note.gnu.property .note.gnu.build-id .interp .gnu.hash .dynsym .dynstr .gnu.version .gnu.version_r .rela.dyn .rela.plt
03 .init .plt .plt.got .text .fini
04 .rodata .eh_frame_hdr .eh_frame .note.ABI-tag
05 .init_array .fini_array .dynamic .got .got.plt .data .bss
06 .dynamic
07 .note.gnu.property
```

- Disassemble

```

rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek) ~/Desktop/filetugas/sistem_operasi
$ objdump -d program | head -50

program:      file format elf64-x86-64

Disassembly of section .init:

0000000000001000 <.init>:
1000: 48 83 ec 08      sub    $0x8,%rsp
1004: 48 05 c5 2f 00 00 mov    0x2fc5(%rip),%rax      # 3fd0 <__gmon_start__@Base>
100b: 48 85 c0         test   %rax,%rax
100e: 74 02          je     1012 <.init+0x12>
1010: ff d0         call   %rax
1012: 48 83 c4 08      add    $0x8,%rsp
1016: c3             ret

Disassembly of section .plt:

0000000000001020 <printf@plt+0x10>:
1020: ff 35 ca 2f 00 00 jmp     0x2fca(%rip)      # 3ff0 <GLOBAL_OFFSET_TABLE+0x8>
1026: ff 25 cc 2f 00 00 jmp     *0x2fcc(%rip)      # 3ff8 <GLOBAL_OFFSET_TABLE+0x10>
102c: 0f 1f 40 00      nopl   0x0(%rax)

0000000000001030 <printf@plt>:
1030: ff 35 ca 2f 00 00 jmp     *0x2fca(%rip)      # 4000 <printf@GLIBC_2.2.5>
1036: 68 00 00 00 00 00 push    $0x0
103b: e9 e0 ff ff     jmp     1020 <.init+0x20>

Disassembly of section .plt.got:

0000000000001040 <_cxa_finalize@plt>:
1040: ff 25 9a 2f 00 00 jmp     *0x2f9a(%rip)      # 3fe0 <_cxa_finalize@GLIBC_2.2.5>
1046: 66 90          xchgb  %ax,%ax

Disassembly of section .text:

0000000000001050 <_start>:
1050: 31 ed          xor     %ebp,%ebp
1052: 49 89 d1        mov     %rdx,%r9
1055: 5e             pop     %rsi
1056: 48 89 e2        mov     %rsp,%rdx
1059: 48 83 e4 f0     and     $0xfffffffffffff0,%rsp
105d: 50             push    %rax
105e: 54             push    %rsp
105f: 45 31 c0        xor     %rdi,%rdi
1062: 31 c9          xor     %ecx,%ecx
1064: 48 8d 3d ce 00 00 00 lea     0xce(%rip),%rdi      # 1139 <main>
106b: ff 15 4f 2f 00 00 call    *0x2f4f(%rip)      # 3fc0 <__libc_start_main@GLIBC_2.34>
1071: f4             hlt
1072: 66 2e 0f 1f 84 00 00 cs nopl 0x0(%rax,%rax,1)
```

7. OS DESIGN AND IMPLEMENTATION

7.1. Binary Format Analysis

7.2. Monitoring Kernel

- Lihat informasi kernel

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ uname -r
6.12.13-amd64

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat /proc/version
Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-16) 14.2
.0, GNU ld (GNU Binutils for Debian) 2.44) #1 SMP PREEMPT_DYNAMIC Kali 6.12.13-1kali1 (2025-0
2-11)
```

- Lihat parameter kernel

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.apparmor_display_secid_mode'
abi.vsyscall32 = 1
crypto.fips_enabled = 0
crypto.fips_name = Linux Kernel Cryptographic API
crypto.fips_version = 6.12.13-amd64
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.hpet.max-user-freq = 64
dev.i915.oa_max_sample_rate = 100000
sysctl: dev.i915.perf_stream_paranoid = 1
permission denied on key 'kernel.apparmor_restrict_unprivileged_unconfined'
dev.mac_hid.mouse_
button2_keycode = 97

dev.mac_hid.mouse_button3_keycode = 100
dev.mac_hid.mouse_button_emulation = 0
dev.scsi.logging_level = 0
dev.tty.ldisc_autoload = 1
dev.tty.legacy_tiocsti = 0
fs.aio-max-nr = 65536
fs.aio-nr = 0
fs.binfmt_misc.python3/13 = enabled
fs.binfmt_misc.python3/13 = interpreter /usr/bin/python3.13
fs.binfmt_misc.python3/13 = flags:
```

- Lihat kernel ring buffer

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ dmesg | tail -30
[ 115.939489] audit: type=1400 audit(1760062896.308:188): apparmor="DENIED" operation="open"
class="file" profile="snap.onlyoffice-desktopeditors.onlyoffice-desktopeditors" name="/etc/v
ulkan/implicit_layer.d/" pid=2496 comm="editors_helper" requested_mask="r" denied_mask="r" fs
uid=1000 ouid=0
[ 115.939492] audit: type=1400 audit(1760062896.308:189): apparmor="DENIED" operation="open"
class="file" profile="snap.onlyoffice-desktopeditors.onlyoffice-desktopeditors" name="/etc/v
ulkan/implicit_layer.d/" pid=2496 comm="editors_helper" requested_mask="r" denied_mask="r" fs
uid=1000 ouid=0
[ 115.940023] audit: type=1400 audit(1760062896.312:190): apparmor="DENIED" operation="open"
class="file" profile="snap.onlyoffice-desktopeditors.onlyoffice-desktopeditors" name="/etc/v
ulkan/implicit_layer.d/" pid=2496 comm="editors_helper" requested_mask="r" denied_mask="r" fs
uid=1000 ouid=0
[ 191.847232] toshiba_acpi: Unknown key 401
[ 240.919221] toshiba_acpi: Unknown key 401
[ 289.994580] toshiba_acpi: Unknown key 401
[ 339.073689] toshiba_acpi: Unknown key 401
[ 388.150987] toshiba_acpi: Unknown key 401
[ 437.225085] toshiba_acpi: Unknown key 401
[ 486.300460] toshiba_acpi: Unknown key 401
[ 535.376485] toshiba_acpi: Unknown key 401
[ 584.454303] toshiba_acpi: Unknown key 401
[ 633.530816] toshiba_acpi: Unknown key 401
[ 682.607926] toshiba_acpi: Unknown key 401
[ 731.685233] toshiba_acpi: Unknown key 401
[ 780.765177] toshiba_acpi: Unknown key 401
[ 829.841641] toshiba_acpi: Unknown key 401
[ 878.920437] toshiba_acpi: Unknown key 401
[ 928.000465] toshiba_acpi: Unknown key 401
[ 977.073143] toshiba_acpi: Unknown key 401
[ 1026.152498] toshiba_acpi: Unknown key 401
[ 1075.229189] toshiba_acpi: Unknown key 401
[ 1124.306787] toshiba_acpi: Unknown key 401
[ 1173.384834] toshiba_acpi: Unknown key 401
[ 1222.457696] toshiba_acpi: Unknown key 401
[ 1271.536699] toshiba_acpi: Unknown key 401
[ 1320.612505] toshiba_acpi: Unknown key 401
[ 1369.691289] toshiba_acpi: Unknown key 401
[ 1418.766754] toshiba_acpi: Unknown key 401
[ 1467.839844] toshiba_acpi: Unknown key 401

```

- Lihat kernel modules

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lsmod
Module                               Size  Used by
snd_seq_dummy                        12288  0
snd_hrtimer                          12288  1
snd_seq                             110592  7 snd_seq_dummy
snd_seq_device                       16384  1 snd_seq
ccm                                  20480  6
qrtr                                 57344  2
sunrpc                               872448  1
binfmt_misc                         28672  1
btusb                                77824  0
btrtl                                32768  1 btusb
intel_rapl_msr                       20480  0
intel_rapl_common                    53248  1 intel_rapl_msr
btintel                              69632  1 btusb
x86_pkg_temp_thermal                 16384  0
nls_ascii                            12288  1
intel_powerclamp                     16384  0
btbcm                                 24576  1 btusb
nls_cp437                            16384  1
coretemp                             16384  0
btmtk                                 32768  1 btusb
vfat                                 24576  1
fat                                  102400  1 vfat
kvm_intel                           413696  0
bluetooth                           1093632  6 btrtl,btmtk,btintel,btbcm,btusb
uvcvideo                             151552  0
videobuf2_vmalloc                     20480  1 uvcvideo
uvc                                    12288  1 uvcvideo
videobuf2_memops                      16384  1 videobuf2_vmalloc
videobuf2_v4l2                        36864  1 uvcvideo
kvm                                  1396736  1 kvm_intel
videodev                              368640  2 videobuf2_v4l2,uvcvideo
videobuf2_common                      81920  4 videobuf2_vmalloc,videobuf2_v4l2,uvcvideo,videobuf2_memops
mc                                    94208  4 videodev,videobuf2_v4l2,uvcvideo,videobuf2_common
iwldvm                               258048  0
crct10dif_pclmul                     12288  1
mei_hdcp                             28672  0
ghash_clmulni_intel                  16384  0
mei_pxp                              16384  0
mac80211                             1445888  1 iwldvm
sha512_ssse3                          53248  0
sha1_ssse3                           32768  0
aesni_intel                          122880  4
gf128mul                             16384  1 aesni_intel
libarc4                              12288  1 mac80211
crypto_simd                           16384  1 aesni_intel
snd_hda_codec_hdmi                    98304  1
cryptd                                28672  2 crypto_simd,ghash_clmulni_intel
```

- Informasi modul tertentu


```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ modinfo ext4
filename:          /lib/modules/6.12.13-amd64/kernel/fs/ext4/ext4.ko.xz
softdep:           pre: crypto-crc32c
license:           GPL
description:       Fourth Extended Filesystem
author:            Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and
others
alias:             fs-ext4
alias:             ext3
alias:             fs-ext3
alias:             ext2
alias:             fs-ext2
depends:            jbd2,crc16,mbcache
intree:           Y
name:              ext4
retpoline:        Y
vermagic:         6.12.13-amd64 SMP preempt mod_unload modversions
sig_id:           PKCS#7
signer:           Build time autogenerated kernel key
sig_key:          5C:F9:36:2A:2E:96:88:06:9E:77:CA:A6:75:34:D8:98:C6:FD:66:BF
sig_hashalgo:     sha256
signature:        1E:47:CC:6F:74:80:8C:4C:11:30:72:9B:FC:36:F7:BC:BF:F6:56:64:
2C:F0:DA:42:AF:95:B1:89:8D:21:73:D3:F5:87:2B:7A:AC:EB:78:70:
F8:2B:FF:4A:7F:4B:89:71:41:F6:37:4C:D5:13:B0:0D:31:49:84:77:
FC:98:4D:F5:FB:1F:D9:1B:F7:09:E1:9B:5B:2F:7E:8F:E7:26:29:01:
3A:EE:41:7B:09:95:A3:19:47:E9:18:3F:2C:DB:92:1C:9F:86:A0:82:
6B:B9:35:08:70:CD:25:80:17:4F:16:28:16:7A:A7:9D:C3:EF:66:4F:
5E:0E:50:23:CA:8C:86:D3:89:E5:70:8F:68:00:6F:0F:7F:C1:65:75:
8A:0D:1B:0F:6C:6C:56:FD:D8:D0:BA:92:BB:DC:35:07:98:8E:9C:AE:
CF:25:C6:B7:A3:8C:BF:EE:CD:C2:A4:AB:C4:40:82:4D:D0:F0:36:07:
0E:C1:81:78:85:48:F6:89:61:94:94:9E:EA:95:37:95:3A:C9:2E:3B:
89:54:FB:F4:89:67:93:7B:92:8C:A6:60:B2:DD:B4:5F:EF:F2:D6:EF:
DD:E8:3D:90:41:4E:E2:1B:A4:BC:D6:5C:90:0D:0E:5A:62:72:D8:59:
80:CA:71:0C:6D:EE:03:49:95:E6:E2:60:1A:C9:20:F8:72:BF:DE:04:
41:8B:F4:3E:9F:A7:CB:A4:CA:35:CE:47:AC:ED:7C:A8:9E:7A:B6:E0:
F6:5D:40:5C:A7:57:48:55:9E:49:89:CC:DC:83:23:38:4C:FF:0A:BB:
E5:C4:38:88:0A:47:8B:60:8D:79:9C:32:48:9B:00:10:B8:1A:49:23:
39:DC:82:5E:8D:FB:1C:8B:D6:66:EA:0F:7B:2C:0C:3F:0A:00:6E:0A:
CD:23:E7:E7:AF:53:EB:E2:69:C2:67:9B:7E:A7:A3:CA:DA:96:7F:38:
AB:8E:07:B6:84:EB:BE:EC:99:11:EC:02:97:0F:BD:EF:A3:31:71:44:
5E:C7:20:98:0D:04:69:44:7F:50:4A:E8:FA:EA:3C:F6:53:38:5B:39:
A5:53:77:C0:0D:23:F8:C5:12:01:18:B6:A9:A1:BF:EB:9B:6F:8A:1A:
8C:AD:EF:01:A2:78:9A:2C:89:F3:3D:3F:52:62:0F:FA:39:53:8F:90:
69:66:73:2E:02:37:31:67:C5:1B:37:91:71:1D:B6:0F:AC:5E:6C:82:
35:03:B4:4B:44:9F:45:44:C5:30:7A:6F:10:1C:A2:09:B1:9E:B9:33:
CD:29:E2:F8:BF:78:73:14:F7:6F:B0:7C:BB:4A:88:E5:B1:69:4B:65:
2D:45:84:6E:D1:07:18:91:EC:6E:BC:E4

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

8. OS STRUCTURE

8.1. Eksplorasi Struktur Layering

- Layer 1: Hardware


```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lspci
00:00.0 Host bridge: Intel Corporation 3rd Gen Core processor DRAM Controller (rev 09)
00:02.0 VGA compatible controller: Intel Corporation 3rd Gen Core processor Graphics Controller (rev 09)
00:14.0 USB controller: Intel Corporation 7 Series/C210 Series Chipset Family USB xHCI Host Controller (rev 04)
00:16.0 Communication controller: Intel Corporation 7 Series/C216 Chipset Family MEI Controller #1 (rev 04)
00:19.0 Ethernet controller: Intel Corporation 82579V Gigabit Network Connection (rev 04)
00:1a.0 USB controller: Intel Corporation 7 Series/C216 Chipset Family USB Enhanced Host Controller #2 (rev 04)
00:1b.0 Audio device: Intel Corporation 7 Series/C216 Chipset Family High Definition Audio Controller (rev 04)
00:1c.0 PCI bridge: Intel Corporation 7 Series/C216 Chipset Family PCI Express Root Port 1 (rev c4)
00:1c.2 PCI bridge: Intel Corporation 7 Series/C210 Series Chipset Family PCI Express Root Port 3 (rev c4)
00:1d.0 USB controller: Intel Corporation 7 Series/C216 Chipset Family USB Enhanced Host Controller #1 (rev 04)
00:1f.0 ISA bridge: Intel Corporation HM76 Express Chipset LPC Controller (rev 04)
00:1f.2 SATA controller: Intel Corporation 7 Series Chipset Family 6-port SATA Controller [AHCI mode] (rev 04)
01:00.0 System peripheral: Ricoh Co Ltd PCIe SDXC/MMC Host Controller (rev 04)
02:00.0 Network controller: Intel Corporation Centrino Advanced-N 6235 (rev 24)

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 8087:0024 Intel Corp. Integrated Rate Matching Hub
Bus 001 Device 003: ID 08ff:168b AuthenTec, Inc. Fingerprint Sensor
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 002: ID 8087:0024 Intel Corp. Integrated Rate Matching Hub
Bus 002 Device 003: ID 1bcf:288e Sunplus Innovation Technology Inc. TOSHIBA Web Camera - HD
Bus 002 Device 004: ID 8087:07da Intel Corp. Centrino Bluetooth Wireless Transceiver
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
```

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat /proc/cpuinfo
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 58
model name     : Intel(R) Core(TM) i7-3667U CPU @ 2.00GHz
stepping       : 9
microcode      : 0x21
cpu MHz        : 563.377
cache size     : 4096 KB
physical id    : 0
siblings       : 4
core id        : 0
cpu cores      : 2
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clf
                lush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx rdtscp lm constant_tsc arch_perfmon p
                ebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfmperf pni pclmulqdq dtes64 monitor ds_
                cpl vmx smx est tm2 ssse3 cx16 xtpr pdcm pcid sse4_1 sse4_2 x2apic popcnt tsc_deadline_timer
                aes xsave avx f16c rdrand lahf_lm cpuid_fault epb pti ssbd ibrs ibpb stibp tpr_shadow flexpri
                ority ept vpid fsgsbase smep erms xsaveopt dtherm ida arat pln pts vnmi md_clear flush_l1d
                vmx flags      : vmni preemption_timer invvpid ept_x_only flexpriority tsc_offset vtptr mtf v
                apic ept vpid unrestricted_guest
                bugs          : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_m
                ultihit srbds mmio_unknown
                bogomips      : 4988.74
                clflush size  : 64
                cache_alignm  : 64
                address sizes  : 36 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id      : GenuineIntel
cpu family     : 6
model          : 58
model name     : Intel(R) Core(TM) i7-3667U CPU @ 2.00GHz
stepping       : 9
microcode      : 0x21
cpu MHz        : 374.603
cache size     : 4096 KB
physical id    : 0
siblings       : 4
core id        : 0
cpu cores      : 2
apicid         : 1
```

- Layer 2: Kernel

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ uname -a
Linux breaksek 6.12.13-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.13-1kali1 (2025-02-11) x86_64 GNU/Linux

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat /proc/sys/kernel/ostype
Linux
```

- Layer 3: System Calls

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffe7c54ce80 /* 58 vars */) = 0
brk(NULL) = 0x55de1ec34000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa6bef1c000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=79787, ...}) = 0
mmap(NULL, 79787, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fa6bef08000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0" ... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=202984, ...}) = 0
mmap(NULL, 210640, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fa6beed4000
mmap(0x7fa6beedb000, 131072, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6beedb000) = 0x7fa6beedb000
mmap(0x7fa6beefb000, 36864, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6beefb000) = 0x7fa6beefb000
mmap(0x7fa6bef04000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6bef04000) = 0x7fa6bef04000
mmap(0x7fa6bef06000, 5840, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fa6bef06000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libcap.so.2", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0" ... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=47288, ...}) = 0
mmap(NULL, 45128, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fa6beec8000
mmap(0x7fa6beecb000, 20480, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6beecb000) = 0x7fa6beecb000
mmap(0x7fa6beed0000, 8192, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6beed0000) = 0x7fa6beed0000
mmap(0x7fa6beed2000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6beed2000) = 0x7fa6beed2000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0p\236\2\0\0\0\0\0" ... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0" ... , 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2003408, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0" ... , 840, 64) = 840
mmap(NULL, 2055800, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fa6becd2000
mmap(0x7fa6becfa000, 1462272, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6becfa000) = 0x7fa6becfa000
mmap(0x7fa6bee5f000, 352256, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6bee5f000) = 0x7fa6bee5f000
mmap(0x7fa6beeb5000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fa6beeb5000) = 0x7fa6beeb5000
mmap(0x7fa6beebb000, 52856, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fa6beebb000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0" ... , 832) = 832
```

- Layer 4: System Programs

rif@breaksek: ~/Desktop/filetugas/sistem_operasi

Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]

\$ ps aux

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.2	25044	15320	?	Ss	09:19	0:02	/sbin/init splash
root	2	0.0	0.0	0	0	?	S	09:19	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	S	09:19	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-rcu_gp]
root	5	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-sync_wq]
root	6	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-slub_flushwq]
root	7	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-netns]
root	9	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/0:0H-events_highp]
root	12	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-mm_percpu_wq]
root	13	0.0	0.0	0	0	?	I	09:19	0:00	[rcu_tasks_kthread]
root	14	0.0	0.0	0	0	?	I	09:19	0:00	[rcu_tasks_rude_kthread]
root	15	0.0	0.0	0	0	?	I	09:19	0:00	[rcu_tasks_trace_kthread]
root	16	0.0	0.0	0	0	?	S	09:19	0:00	[ksoftirqd/0]
root	17	0.2	0.0	0	0	?	I	09:19	0:06	[rcu_preempt]
root	18	0.0	0.0	0	0	?	S	09:19	0:00	[rcu_exp_par_gp_kthread_wo]
root	19	0.0	0.0	0	0	?	S	09:19	0:00	[rcu_exp_gp_kthread_worker]
root	20	0.0	0.0	0	0	?	S	09:19	0:00	[migration/0]
root	21	0.0	0.0	0	0	?	S	09:19	0:00	[idle_inject/0]
root	22	0.0	0.0	0	0	?	S	09:19	0:00	[cpuhp/0]
root	23	0.0	0.0	0	0	?	S	09:19	0:00	[cpuhp/2]
root	24	0.0	0.0	0	0	?	S	09:19	0:00	[idle_inject/2]
root	25	0.0	0.0	0	0	?	S	09:19	0:00	[migration/2]
root	26	0.0	0.0	0	0	?	S	09:19	0:00	[ksoftirqd/2]
root	28	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/2:0H-events_highp]
root	29	0.0	0.0	0	0	?	S	09:19	0:00	[cpuhp/1]
root	30	0.0	0.0	0	0	?	S	09:19	0:00	[idle_inject/1]
root	31	0.0	0.0	0	0	?	S	09:19	0:00	[migration/1]
root	32	0.0	0.0	0	0	?	S	09:19	0:00	[ksoftirqd/1]
root	34	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/1:0H-events_highp]
root	35	0.0	0.0	0	0	?	S	09:19	0:00	[cpuhp/3]
root	36	0.0	0.0	0	0	?	S	09:19	0:00	[idle_inject/3]
root	37	0.0	0.0	0	0	?	S	09:19	0:00	[migration/3]
root	38	0.0	0.0	0	0	?	S	09:19	0:00	[ksoftirqd/3]
root	40	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/3:0H-events_highp]
root	44	0.0	0.0	0	0	?	I	09:19	0:01	[kworker/u16:3-events_unbo]
root	45	0.0	0.0	0	0	?	S	09:19	0:00	[kdevtmpfs]
root	46	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-inet_frag_wq]
root	47	0.0	0.0	0	0	?	S	09:19	0:00	[kauditd]
root	48	0.0	0.0	0	0	?	S	09:19	0:00	[khungtaskd]
root	49	0.0	0.0	0	0	?	S	09:19	0:00	[oom_reaper]
root	50	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-writeback]
root	51	0.0	0.0	0	0	?	S	09:19	0:00	[kcompactd0]
root	52	0.0	0.0	0	0	?	SN	09:19	0:00	[ksmd]
root	53	0.0	0.0	0	0	?	SN	09:19	0:00	[khugepaged]
root	54	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-kintegrityd]
root	55	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-kblockd]
root	56	0.0	0.0	0	0	?	I<	09:19	0:00	[kworker/R-blkcg_punt_bio]
root	57	0.0	0.0	0	0	?	S	09:19	0:00	[irq/9-acpi]


```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ top
top - 10:05:45 up 46 min, 1 user, load average: 1.31, 1.23, 1.18
Tasks: 225 total, 1 running, 224 sleeping, 0 stopped, 0 zombie
%Cpu(s): 25.2 us, 8.7 sy, 0.0 ni, 66.0 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 5825.8 total, 378.4 free, 2440.1 used, 3533.6 buff/cache
MiB Swap: 6038.0 total, 6038.0 free, 0.0 used, 3385.7 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 1429 root        20   0   680392 144816  89044 S   38.0   2.4   6:21.63 Xorg
 2571 rif         20   0    56.7g 351148 104120 S   21.3   5.9  15:42.93 editors_helper
 4617 rif         20   0   424296  37844  30056 S   17.7   0.6   0:00.54 xfce4-screensho
 2359 rif         20   0  2976212 260492 197632 S   16.1   4.4   8:33.17 DesktopEditors
 2054 rif         20   0   432948  48404  34552 S   13.1   0.8   0:07.49 wrapper-2.0
 2003 rif         20   0   658028 103780  76820 S    6.9   1.7   0:33.95 xfwm4
 2496 rif         20   0  1366556 134968 106016 S    6.2   2.3   3:58.67 editors_helper
 2036 rif         20   0   509916  46796  32332 S    5.2   0.8   0:24.49 xfce4-panel
 3130 rif         20   0   797540  59572  50092 S    1.6   1.0   1:13.88 qterminal
 4609 rif         20   0   10824   5764   3716 R    1.6   0.1   0:00.17 top
 2112 rif         20   0   281524  34700  18404 S    1.3   0.6   0:01.52 xfce4-screensav
 2600 rif         20   0    32.7g 337996 231192 S    1.3   5.7   2:20.30 chrome
 3683 rif         20   0   804044  66120  48848 S    1.3   1.1   0:11.38 qterminal
 1885 rif         20   0   422236  36164  18860 S    1.0   0.6   0:01.97 xfce4-session
 2049 rif         20   0   508728  59224  38460 S    1.0   1.0   0:02.17 xfdesktop
   17 root         20   0         0         0         0 I    0.3   0.0   0:06.45 rcu_preempt
   44 root         20   0         0         0         0 I    0.3   0.0   0:01.93 kworker/u16:3-iwlwifi
   58 root         20   0         0         0         0 I    0.3   0.0   0:01.44 kworker/3:1-events
   59 root         20   0         0         0         0 I    0.3   0.0   0:04.88 kworker/2:1-mm_percp+
 1866 rif         20   0    9084   5468   4060 S    0.3   0.1   0:00.49 dbus-daemon
 1973 rif         20   0    8520   4532   4020 S    0.3   0.1   0:00.37 dbus-daemon
 2026 rif         20   0   361568  40536 19396 S    0.3   0.7   0:02.16 xfsettingsd
 2060 rif         20   0   358888  38972  30824 S    0.3   0.7   0:00.79 wrapper-2.0
 2084 rif         20   0   500160  38428  30496 S    0.3   0.6   0:00.40 xfce4-notifyd
 2100 rif         20   0   405588  18644  16212 S    0.3   0.3   0:00.26 xfce4-power-man
 2117 rif         20   0   603644  59660  33796 S    0.3   1.0   0:00.69 bluedevil-applet
 2381 rif         20   0   540084   6820   6308 S    0.3   0.1   0:00.06 xdg-document-po
 2642 rif         20   0    32.8g 212820 137548 S    0.3   3.6   1:27.39 chrome
 3646 _apt         20   0   25048  14744 12056 S    0.3   0.2   0:08.63 https
 4248 root          0 -20         0         0         0 I    0.3   0.0   0:01.93 kworker/u17:0-i915_f+
 4488 root         20   0         0         0         0 I    0.3   0.0   0:00.48 kworker/u16:2-iwlwifi
 4516 rif         20   0   308072   6320   5936 S    0.3   0.1   0:00.04 xfconfd
    1 root         20   0   25044  15320 10820 S    0.0   0.3   0:02.37 systemd
    2 root         20   0         0         0         0 S    0.0   0.0   0:00.00 kthreadd
    3 root         20   0         0         0         0 S    0.0   0.0   0:00.00 pool_workqueue_relea+
    4 root          0 -20         0         0         0 I    0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root          0 -20         0         0         0 I    0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root          0 -20         0         0         0 I    0.0   0.0   0:00.00 kworker/R-slub_flush+
    7 root          0 -20         0         0         0 I    0.0   0.0   0:00.00 kworker/R-netns
    9 root          0 -20         0         0         0 I    0.0   0.0   0:00.00 kworker/0:0H-events_+
   12 root          0 -20         0         0         0 I    0.0   0.0   0:00.00 kworker/R-mm_percpu_+
```

- Layer 5: User Applications

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ which bash
/usr/bin/bash

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ which gcc
/usr/bin/gcc
```

8.2. Microkernel vs Monolithic

- Lihat ukuran kernel (Monolithic)

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ls -lh /boot/vmlinuz-$(uname -r)
-rw-r--r-- 1 root root 9.6M Feb 11 2025 /boot/vmlinuz-6.12.13-amd64
```

- Lihat built-in vs module

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
2638

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat /boot/config-$(uname -r) | grep "=m" | wc -l
4007
```

- Lihat loaded modules

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lsmod | wc -l
133

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

8.3. Program - Virtual File System

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo nano procfs_read.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -o procfs_read procfs_read.c

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ./procfs_read
=== CPU Information ===
processor      : 0
vendor_id     : GenuineIntel
cpu family    : 6
model         : 58
model name    : Intel(R) Core(TM) i7-3667U CPU @ 2.00GHz
stepping      : 9
microcode     : 0x21
cpu MHz       : 333.678
cache size    : 4096 KB
physical id   : 0

=== Memory Information ===
MemTotal:     5965592 kB
MemFree:      688504 kB
MemAvailable: 3366332 kB
Buffers:      82124 kB
Cached:       2964708 kB

=== Process Information ===
Name:   procfs_read
Umask:  0002
State:  R (running)
Tgid:   6230
Ngid:   0
Pid:    6230
PPid:   3690
TracerPid: 0
Uid:    1000 1000 1000 1000
Gid:    1000 1000 1000 1000
```

9. BUILDING AND BOOTING OS

9.1. Analisis Boot Process

- Lihat boot messages


```

rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help
[ 0.000000] Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-16) 14.2.0, GNU ld (GNU Binutils for Debian) 2.44) #1 SMP PREEMPT_DYNAMIC Kali 6.12.13-1kal
i1 (2025-02-11)
[ 0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=64522bb6-073a-4e61-8906-184df2531bf8 ro quiet splash
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000097fff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000098000-0x00000000000bffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000020000000-0x00000000201fffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000020200000-0x000000004003ffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000040040000-0x000000004004ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000040050000-0x00000000ca28ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000ca28f000-0x00000000ca28ffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x00000000ca290000-0x00000000ca291fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca292000-0x00000000ca34ffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x00000000ca350000-0x00000000ca352fff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x00000000ca353000-0x00000000ca38ffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x00000000ca38f000-0x00000000ca3aefff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x00000000ca3af000-0x00000000ca3affff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3b0000-0x00000000ca3bffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000ca3c0000-0x00000000ca3cffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3d0000-0x00000000ca3dffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3e0000-0x00000000ca3effff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3f0000-0x00000000ca3f1fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3f2000-0x00000000ca3f3fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3f4000-0x00000000ca3f5fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3f6000-0x00000000ca3f7fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3f8000-0x00000000ca3f9fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3fa000-0x00000000ca3fbfff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3fc000-0x00000000ca3fdfff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3fe000-0x00000000ca3fefff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000ca3ff000-0x00000000ca3fffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] e820: update [mem 0xc7dc7018-0xc7de7057] usable ==> usable
[ 0.000000] e820: update [mem 0xc7dc8018-0xc7dd6e57] usable ==> usable
[ 0.000000] extended physical RAM map:
[ 0.000000] reserve setup_data: [mem 0x0000000000000000-0x0000000000097fff] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000098000-0x00000000000bffff] reserved
[ 0.000000] reserve setup_data: [mem 0x0000000000100000-0x00000000001fffff] usable
[ 0.000000] reserve setup_data: [mem 0x0000000020000000-0x00000000201fffff] reserved
[ 0.000000] reserve setup_data: [mem 0x0000000020200000-0x000000004003ffff] usable
[ 0.000000] reserve setup_data: [mem 0x0000000040040000-0x000000004004ffff] reserved
[ 0.000000] reserve setup_data: [mem 0x0000000040050000-0x000000004004ffff] reserved
[ 0.000000] reserve setup_data: [mem 0x0000000040050000-0x00000000c7dc8017] usable
[ 0.000000] reserve setup_data: [mem 0x00000000c7dc8018-0x00000000c7dd6e57] usable
[ 0.000000] reserve setup_data: [mem 0x00000000c7dd6e58-0x00000000c7dd7017] usable
[ 0.000000] reserve setup_data: [mem 0x00000000c7dd7018-0x00000000c7de7057] usable
[ 0.000000] reserve setup_data: [mem 0x00000000c7de7058-0x00000000ca28ffff] usable
[ 0.000000] reserve setup_data: [mem 0x00000000ca28f000-0x00000000ca28ffff] ACPI NVS
[ 0.000000] reserve setup_data: [mem 0x00000000ca290000-0x00000000ca291fff] reserved
[ 0.000000] reserve setup_data: [mem 0x00000000ca292000-0x00000000ca34ffff] ACPI NVS
[ 0.000000] reserve setup_data: [mem 0x00000000ca350000-0x00000000ca352fff] ACPI data
[ 0.000000] reserve setup_data: [mem 0x00000000ca353000-0x00000000ca38ffff] ACPI NVS
[ 0.000000] reserve setup_data: [mem 0x00000000ca38f000-0x00000000ca3aefff] ACPI data
[ 0.000000] reserve setup_data: [mem 0x00000000ca38f000-0x00000000ca3aefff] ACPI data

```

- Lihat systemd boot analysis

```

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ systemd-analyze
Startup finished in 4.559s (firmware) + 10.754s (loader) + 4.780s (kernel) + 14.579s (userspace) = 34.674s
graphical.target reached after 7.779s in userspace.

```

- Lihat GRUB configuration

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo cat /boot/grub/grub.cfg | head -50
[sudo] password for rif:
#
# DO NOT EDIT THIS FILE
#
# It is automatically generated by grub-mkconfig using templates
# from /etc/grub.d and settings from /etc/default/grub
#

### BEGIN /etc/grub.d/00_header ###
if [ -s $prefix/grubenv ]; then
    set have_grubenv=true
    load_env
fi
if [ "${next_entry}" ]; then
    set default="${next_entry}"
    set next_entry=
    save_env next_entry
    set boot_once=true
else
    set default="0"
fi

if [ x"${feature_menuentry_id}" = xy ]; then
    menuentry_id_option="--id"
else
    menuentry_id_option=""
fi

export menuentry_id_option

if [ "${prev_saved_entry}" ]; then
    set saved_entry="${prev_saved_entry}"
    save_env saved_entry
    set prev_saved_entry=
    save_env prev_saved_entry
    set boot_once=true
fi

function savedefault {
    if [ -z "${boot_once}" ]; then
        saved_entry="${chosen}"
        save_env saved_entry
    fi
}
function load_video {
    if [ x$feature_all_video_module = xy ]; then
        insmod all_video
    else
        insmod efi_gop
```

- Lihat initramfs

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
kernel
kernel/x86
kernel/x86/microcode
kernel/x86/microcode/.enuineIntel.align.0123456789abc
kernel/x86/microcode/GenuineIntel.bin
.
usr
usr/lib
usr/lib/modules
usr/lib/modules/6.12.13-amd64
usr/lib/modules/6.12.13-amd64/kernel
usr/lib/modules/6.12.13-amd64/kernel/arch
usr/lib/modules/6.12.13-amd64/kernel/arch/x86
usr/lib/modules/6.12.13-amd64/kernel/arch/x86/crypto
usr/lib/modules/6.12.13-amd64/kernel/arch/x86/crypto/crc32-pclmul.ko.xz
usr/lib/modules/6.12.13-amd64/kernel/arch/x86/crypto/crc32c-intel.ko.xz
usr/lib/modules/6.12.13-amd64/kernel/arch/x86/crypto/sha256-ssse3.ko.xz
usr/lib/modules/6.12.13-amd64/kernel/arch/x86/kvm
usr/lib/modules/6.12.13-amd64/kernel/arch/x86/kvm/kvm.ko.xz
usr/lib/modules/6.12.13-amd64/kernel/crypto
```

9.2. Kernel Parameters

- Lihat kernel parameters saat boot

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat /proc/cmdline
BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=64522bb6-073a-4e61-8906-184df2531bf8 ro quiet splash
```

9.3. Simple Bootloader (Konsep)

10. OS DEBUGGING

10.1. Using strace

- Trace system calls


```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ strace -c ls
custom_daemon      info_syscall.c  math_ops.c      pelayanan_os     program_static
custom_daemon.c    libmathops.so  math_ops.h      pelayanan_os.c   proses_syscall
detail_os.c        main.c         math_ops.i      procfs_read      proses_syscall.c
file_syscall       main.i         math_ops.o      procfs_read.c    shell_custom
file_syscall.c     main.o         math_ops.s      program          shell_custom.c
hallo_module.c     main.s         output.tx       program_dynamic  syscall_test.txt
info_syscall       Makefile       output.txt      program_shared   test_file.txt
% time      seconds  usecs/call   calls    errors syscall
-----
29.96      0.000432    19          22        mmap
13.18      0.000190    31           6        mprotect
 8.95      0.000129    16           8        openat
 7.42      0.000107    15           7        write
 6.31      0.000091     9          10        close
 4.79      0.000069    11           6        read
 4.37      0.000063    31           2        getdents64
 4.16      0.000060     6           9        fstat
 3.95      0.000057     9           6        4 prctl
 3.81      0.000055    55           1        munmap
 2.50      0.000036    18           2        2 statfs
 2.36      0.000034    17           2        pread64
 1.80      0.000026     8           3        brk
 1.32      0.000019     9           2        ioctl
 0.83      0.000012    12           1        arch_prctl
 0.83      0.000012    12           1        prlimit64
 0.76      0.000011     5           2        2 access
 0.76      0.000011    11           1        rseq
 0.69      0.000010    10           1        set_tid_address
 0.62      0.000009     9           1        set_robust_list
 0.62      0.000009     9           1        getrandom
 0.00      0.000000     0           1        execve
-----
100.00     0.001442    15          95        8 total
```

- Trace dan simpan ke file


```

rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ strace -o trace.txt ls
custom_daemon    libmathops.so    math_ops.i       procfs_read.c    shell_custom.c
custom_daemon.c  main.c           math_ops.o       program           syscall_test.txt
detail_os.c      main.i           math_ops.s       program_dynamic   test_file.txt
file_syscall     main.o           output.tx        program_shared    trace.txt
file_syscall.c   main.s           output.txt       program_static
hallo_module.c   Makefile         pelayananan_os   proses_syscall
info_syscall     math_ops.c       pelayananan_os.c proses_syscall.c
info_syscall.c   math_ops.h       procfs_read      shell_custom

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat trace.txt
execve("/usr/bin/ls", ["ls"], 0x7ffce6b07b10 /* 58 vars */) = 0
brk(NULL) = 0x5611f4f11000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fbb7ee3b000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=79787, ...}) = 0
mmap(NULL, 79787, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fbb7ee27000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0" ... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=202984, ...}) = 0
mmap(NULL, 210640, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fbb7edf3000
mmap(0x7fbb7edfa000, 131072, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fbb7edfa000) = 0x7fbb7edfa000
mmap(0x7fbb7ee1a000, 36864, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fbb7ee1a000) = 0x7fbb7ee1a000
mmap(0x7fbb7ee23000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fbb7ee23000) = 0x7fbb7ee23000
mmap(0x7fbb7ee25000, 5840, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fbb7ee25000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libcap.so.2", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0" ... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=47288, ...}) = 0
mmap(NULL, 45128, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fbb7ede7000
mmap(0x7fbb7edea000, 20480, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fbb7edea000) = 0x7fbb7edea000
mmap(0x7fbb7edef000, 8192, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fbb7edef000) = 0x7fbb7edef000
mmap(0x7fbb7edf1000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7fbb7edf1000) = 0x7fbb7edf1000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0p\236\2\0\0\0\0\0" ... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0" ... , 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2003408, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0" ... , 840, 64) = 840
mmap(NULL, 2055800, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fbb7ebf1000
```

- 10.2. Using ltrace
- Trace library calls

[illegible]

- Dengan detail

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ ltrace -c ls
custom_daemon    libmathops.so    math_ops.i       procfs_read.c    shell_custom.c
custom_daemon.c  main.c           math_ops.o       program           syscall_test.txt
detail_os.c      main.i           math_ops.s       program_dynamic   test_file.txt
file_syscall     main.o           output.tx        program_shared    trace.txt
file_syscall.c   main.s           output.txt       program_static
hallo_module.c   Makefile         pelayananan_os   proses_syscall
info_syscall     math_ops.c       pelayananan_os.c proses_syscall.c
info_syscall.c   math_ops.h       procfs_read      shell_custom

% time      seconds    usecs/call      calls      function
-----
27.23      0.049162      261             188      __errno_location
23.91      0.043169      289             149      strcoll
14.81      0.026740      184             145      strlen
10.26      0.018519      171             108      __ctype_get_mb_cur_max
6.32       0.011417      200             57       __overflow
6.07       0.010959      192             57       memcpy
3.89       0.007020      195             36       fwrite_unlocked
3.70       0.006680      171             39       readdir
0.65       0.001165      145             8         getenv
0.49       0.000893      297             3         realloc
0.40       0.000727      727             1         setlocale
0.30       0.000546      136             4         __freading
0.20       0.000370      185             2         fclose
0.19       0.000343      343             1         exit_group
0.17       0.000315      315             1         strrchr
0.17       0.000314      157             2         __fpending
0.16       0.000294      147             2         fflush
0.16       0.000283      283             1         opendir
0.15       0.000274      137             2         fileno
0.12       0.000214      214             1         isatty
0.11       0.000198      198             1         closedir
0.11       0.000192      192             1         bindtextdomain
0.10       0.000182      182             1         ioctl
0.09       0.000171      171             1         _setjmp
0.08       0.000147      147             1         getopt_long
0.08       0.000140      140             1         textdomain
0.08       0.000138      138             1         __cxa_atexit

100.00     0.180572                        814      total

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

10.3. Using GDB

- Compile dengan debug symbols

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -g -o buggy buggy.c
```

- Jalankan dengan GDB

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gdb ./buggy
GNU gdb (Debian 16.3-5) 16.3
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word" ...
Reading symbols from ./buggy ...
(gdb)
```


- Di dalam GDB

```

rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gdb ./buggy
GNU gdb (Debian 16.3-5) 16.3
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word" ...
Reading symbols from ./buggy ...
(gdb) break main
Breakpoint 1 at 0x1200: file buggy.c, line 17.
(gdb) run
Starting program: /home/rif/Desktop/filetugas/sistem_operasi/buggy
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:17
17      buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
18      return 0;
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 9217) exited normally]
(gdb) backtrace
No stack.
(gdb) quit

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ █

```

10.4. Kernel Debugging dengan dmesg

- Lihat kernel log

```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ dmesg
[ 0.000000] Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian
14.2.0-16) 14.2.0, GNU ld (GNU Binutils for Debian) 2.44) #1 SMP PREEMPT_DYNAMIC Kali 6.12.13
-1kali1 (2025-02-11)
[ 0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=64522bb6-073a-4
e61-8906-184df2531bf8 ro quiet splash
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x00000000000097fff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000098000-0x000000000000bffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000100000-0x000000000001fffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000200000-0x00000000000201ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000202000-0x0000000000040003fff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000040004000-0x0000000000040004fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000040005000-0x00000000000ca28efff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000ca28f000-0x00000000000ca28ffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x00000000000ca290000-0x00000000000ca291fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000ca292000-0x00000000000ca34ffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x00000000000ca350000-0x00000000000ca352fff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x00000000000ca353000-0x00000000000ca38efff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x00000000000ca38f000-0x00000000000ca3aefff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x00000000000ca3af000-0x00000000000caffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000cafff000-0x00000000000caffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000cb000000-0x00000000000cf9ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000f8000000-0x00000000000fbffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000fec00000-0x00000000000fec0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000fed10000-0x00000000000fed13fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000fed18000-0x00000000000fed19fff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000fed1c000-0x00000000000fed1ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000fee00000-0x00000000000fee0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000ffc00000-0x00000000000ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000010000000-0x000000000001af5ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] e820: update [mem 0xc7dd7018-0xc7de7057] usable ==> usable
[ 0.000000] e820: update [mem 0xc7dc8018-0xc7dd6e57] usable ==> usable
[ 0.000000] extended physical RAM map:
[ 0.000000] reserve setup_data: [mem 0x00000000000000000-0x00000000000097fff] usable
[ 0.000000] reserve setup_data: [mem 0x00000000000098000-0x000000000000bffff] reserved
[ 0.000000] reserve setup_data: [mem 0x00000000000100000-0x000000000001fffff] usable
[ 0.000000] reserve setup_data: [mem 0x00000000000200000-0x00000000000201ffff] reserved
[ 0.000000] reserve setup_data: [mem 0x00000000000202000-0x0000000000040003fff] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000040004000-0x0000000000040004fff] reserved
[ 0.000000] reserve setup_data: [mem 0x0000000000040005000-0x00000000000c7dc8017] usable
[ 0.000000] reserve setup_data: [mem 0x00000000000c7dc8018-0x00000000000c7dd6e57] usable
[ 0.000000] reserve setup_data: [mem 0x00000000000c7dd6e58-0x00000000000c7dd7017] usable
[ 0.000000] reserve setup_data: [mem 0x00000000000c7dd7018-0x00000000000c7de7057] usable
[ 0.000000] reserve setup_data: [mem 0x00000000000c7de7058-0x00000000000ca28efff] usable
[ 0.000000] reserve setup_data: [mem 0x00000000000ca28f000-0x00000000000ca28ffff] ACPI NVS
[ 0.000000] reserve setup_data: [mem 0x00000000000ca290000-0x00000000000ca291fff] reserved
```

- Follow kernel log


```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ dmesg -w
[ 0.000000] Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian
14.2.0-16) 14.2.0, GNU ld (GNU Binutils for Debian) 2.44) #1 SMP PREEMPT_DYNAMIC Kali 6.12.13
-1kali1 (2025-02-11)
[ 0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=64522bb6-073a-4
e61-8906-184df2531bf8 ro quiet splash
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000097fff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000098000-0x000000000000bffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] usable
[ 0.000000] BIOS-e820: [mem 0x000000000020000000-0x0000000000201fffff] reserved
[ 0.000000] BIOS-e820: [mem 0x000000000020200000-0x000000000040003fff] usable
[ 0.000000] BIOS-e820: [mem 0x000000000040004000-0x000000000040004fff] reserved
[ 0.000000] BIOS-e820: [mem 0x000000000040005000-0x0000000000ca28efff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000ca28f000-0x0000000000ca28ffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x0000000000ca290000-0x0000000000ca291fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000ca292000-0x0000000000ca34ffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x0000000000ca350000-0x0000000000ca352fff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000ca353000-0x0000000000ca38efff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x0000000000ca38f000-0x0000000000ca3aefff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000ca3af000-0x0000000000caffffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000cafff000-0x0000000000caffffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000cb000000-0x0000000000cf9fffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000f8000000-0x0000000000fbfffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000fec00000-0x0000000000fec0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000fed10000-0x0000000000fed13fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000fed18000-0x0000000000fed19fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000fed1c000-0x0000000000fed1ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000fee00000-0x0000000000fee0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000ffc00000-0x0000000000ffffff] reserved
[ 0.000000] BIOS-e820: [mem 0x000000001000000000-0x000000001af5fffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] e820: update [mem 0xc7dd7018-0xc7de7057] usable ==> usable
[ 0.000000] e820: update [mem 0xc7dc8018-0xc7dd6e57] usable ==> usable
[ 0.000000] extended physical RAM map:
[ 0.000000] reserve setup_data: [mem 0x0000000000000000-0x0000000000097fff] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000098000-0x000000000000bffff] reserved
[ 0.000000] reserve setup_data: [mem 0x0000000000100000-0x00000000001fffff] usable
[ 0.000000] reserve setup_data: [mem 0x000000000020000000-0x0000000000201fffff] reserved
[ 0.000000] reserve setup_data: [mem 0x000000000020200000-0x000000000040003fff] usable
[ 0.000000] reserve setup_data: [mem 0x000000000040004000-0x000000000040004fff] reserved
[ 0.000000] reserve setup_data: [mem 0x000000000040005000-0x0000000000c7dc8017] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000c7dc8018-0x0000000000c7dd6e57] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000c7dd6e58-0x0000000000c7dd7017] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000c7dd7018-0x0000000000c7de7057] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000c7de7058-0x0000000000ca28efff] usable
[ 0.000000] reserve setup_data: [mem 0x0000000000ca28f000-0x0000000000ca28ffff] ACPI NVS
[ 0.000000] reserve setup_data: [mem 0x0000000000ca290000-0x0000000000ca291fff] reserved
[ 0.000000] reserve setup_data: [mem 0x0000000000ca292000-0x0000000000ca34ffff] ACPI NVS
```

- Filter by level


```
rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ dmesg -l err
[ 8.102053] Bluetooth: hci0: unexpected event for opcode 0x0000

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ dmesg -l warn
[ 0.188781] MDS CPU bug present and SMT on, data leak possible. See https://www.kernel.org/doc/html/latest/admin-guide/hw-vuln/mds.html for more details.
[ 0.199865] Simple Boot Flag value 0x0 read from CMOS RAM was invalid
[ 1.052721] ENERGY_PERF_BIAS: Set to 'normal', was 'performance'
[ 1.567183] ACPI Warning: SystemIO range 0x0000000000000428-0x000000000000042F conflicts with OpRegion 0x0000000000000400-0x000000000000047F (\PMIO) (20240827/utaddress-204)
[ 1.567201] ACPI Warning: SystemIO range 0x0000000000000540-0x000000000000054F conflicts with OpRegion 0x0000000000000500-0x000000000000057B (\GPIO) (20240827/utaddress-204)
[ 1.567208] ACPI Warning: SystemIO range 0x0000000000000530-0x000000000000053F conflicts with OpRegion 0x0000000000000500-0x000000000000057B (\GPIO) (20240827/utaddress-204)
[ 1.567215] ACPI Warning: SystemIO range 0x0000000000000500-0x000000000000052F conflicts with OpRegion 0x0000000000000500-0x000000000000057B (\GPIO) (20240827/utaddress-204)
[ 1.567221] lp0:ich: Resource conflict(s) found affecting gpio_ich
[ 7.311473] resource: resource sanity check: requesting [mem 0x00000000fed10000-0x00000000fed15fff], which spans more than Reserved [mem 0xfed10000-0xfed13fff]
[ 7.311482] caller snb_uncore_imc_init_box+0x87/0xd0 [intel_uncore] mapping multiple BARs
[ 7.413471] iwlmwifi 0000:02:00.0: can't disable ASPM; OS doesn't have ASPM control
[ 13.522101] kauditd_printk_skb: 167 callbacks suppressed
[ 25.942150] warning: 'wrapper-2.0' uses wireless extensions which will stop working for Wi-Fi 7 hardware; use nl80211
[11071.373291] kauditd_printk_skb: 13 callbacks suppressed
[12905.744369] hrtimer: interrupt took 228912 ns

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$
```

10.5. Memory Leak Detection dengan Valgrind

- Install valgrind

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo apt install valgrind
valgrind is already the newest version (1:3.25.1-2).
The following packages were automatically installed and are no longer required:
  firmware-ti-connectivity g++-14 g++-14-x86-64-linux-gnu libstdc++-14-dev
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 7
```

- Buat program dengan memory leak

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ cat > memleak.c << 'EOF'
#include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
    return 0;
}
EOF
```

- Compile Program

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ gcc -o memleak memleak.c
```

- Jalankan dengan valgrind

```
(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ valgrind --leak-check=full ./memleak
==9565== Memcheck, a memory error detector
==9565== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==9565== Using Valgrind-3.25.1 and LibVEX; rerun with -h for copyright info
==9565== Command: ./memleak
==9565==
==9565==
==9565== HEAP SUMMARY:
==9565==   in use at exit: 400 bytes in 1 blocks
==9565==   total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==9565==
==9565== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==9565==    at 0x4849818: malloc (vg_replace_malloc.c:446)
==9565==    by 0x400114A: main (in /home/rif/Desktop/filetugas/sistem_operasi/memleak)
==9565==
==9565== LEAK SUMMARY:
==9565==   definitely lost: 400 bytes in 1 blocks
==9565==   indirectly lost: 0 bytes in 0 blocks
==9565==   possibly lost: 0 bytes in 0 blocks
==9565==   still reachable: 0 bytes in 0 blocks
==9565==   suppressed: 0 bytes in 0 blocks
==9565==
==9565== For lists of detected and suppressed errors, rerun with: -s
==9565== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
```

10.6. Performance Analysis

- CPU profiling dengan perf

- Monitor system calls

```

rif@breaksek: ~/Desktop/filetugas/sistem_operasi
Session Actions Edit View Help

(venv)(rif@breaksek)-[~/Desktop/filetugas/sistem_operasi]
$ sudo perf trace ls
buggy          info_syscall  math_ops.c    output.txt     program_static
buggy.c        info_syscall.c math_ops.h     pelayananan_os proses_syscall
custom_daemon  libmathops.so  math_ops.i     pelayananan_os.c proses_syscall.c
custom_daemon.c main.c         math_ops.o     procfs_read     shell_custom
detail_os.c    main.i         math_ops.s     procfs_read.c   shell_custom.c
file_syscall   main.o         memleak        program          syscall_test.txt
file_syscall.c main.s         memleak.c      program_dynamic  test_file.txt
hallo_module.c Makefile       output.tx      program_shared   trace.txt
? (           ): ls/9687 ... [continued]: execve()
= 0
0.128 ( 0.007 ms): ls/9687 brk()
= 0x5634ab75b000
0.239 ( 0.021 ms): ls/9687 mmap(len: 8192, prot: READ|WRITE, flags: PRIVATE|ANONYMOUS)
= 0x7fb8827fe000
0.283 ( 0.010 ms): ls/9687 access(filename: 0x82832480, mode: R)
= -1 ENOENT (No such file or directory)
0.312 ( 0.019 ms): ls/9687 openat(dfd: CWD, filename: 0x8283121c, flags: RDONLY|CLOEXEC)
= 3
0.338 ( 0.007 ms): ls/9687 fstat(fd: 3, statbuf: 0x7ffe587c5df0)
= 0
0.350 ( 0.020 ms): ls/9687 mmap(len: 79787, prot: READ, flags: PRIVATE, fd: 3)
= 0x7fb8827ea000
0.376 ( 0.004 ms): ls/9687 close(fd: 3)
= 0
0.410 ( 0.016 ms): ls/9687 openat(dfd: CWD, filename: 0x827fe140, flags: RDONLY|CLOEXEC)
= 3
0.431 ( 0.007 ms): ls/9687 read(fd: 3, buf: 0x7ffe587c5f68, count: 832)
= 832
0.443 ( 0.004 ms): ls/9687 fstat(fd: 3, statbuf: 0x7ffe587c5df0)
= 0
0.454 ( 0.019 ms): ls/9687 mmap(len: 210640, prot: READ, flags: PRIVATE|DENYWRITE, fd: 3)
= 0x7fb8827b6000
0.479 ( 0.040 ms): ls/9687 mmap(addr: 0x7fb8827bd000, len: 131072, prot: READ|EXEC, flags: PRIVATE|FIXED|DENYWRITE, fd: 3, off: 0x7000) = 0x7fb8827bd000
0.525 ( 0.026 ms): ls/9687 mmap(addr: 0x7fb8827dd000, len: 36864, prot: READ, flags: PRIVATE|FIXED|DENYWRITE, fd: 3, off: 0x27000) = 0x7fb8827dd000
0.557 ( 0.029 ms): ls/9687 mmap(addr: 0x7fb8827e6000, len: 8192, prot: READ|WRITE, flags: PRIVATE|FIXED|DENYWRITE, fd: 3, off: 0x30000) = 0x7fb8827e6000
0.606 ( 0.020 ms): ls/9687 mmap(addr: 0x7fb8827e8000, len: 5840, prot: READ|WRITE, flags: PRIVATE|FIXED|ANONYMOUS) = 0x7fb8827e8000
0.655 ( 0.004 ms): ls/9687 close(fd: 3)
= 0
0.672 ( 0.020 ms): ls/9687 openat(dfd: CWD, filename: 0x827fe690, flags: RDONLY|CLOEXEC)
= 3
0.698 ( 0.007 ms): ls/9687 read(fd: 3, buf: 0x7ffe587c5f48, count: 832)
= 832
0.709 ( 0.005 ms): ls/9687 fstat(fd: 3, statbuf: 0x7ffe587c5dd0)
= 0
0.722 ( 0.032 ms): ls/9687 mmap(len: 45128, prot: READ, flags: PRIVATE|DENYWRITE, fd: 3)

```

- Top-like interface

rif@breaksek: ~/Desktop/filetugas/sistem_operasi

Session Actions Edit View Help

Samples: 63 of event 'cycles:P', 4000 Hz, Event count (approx.): 1245324 lost: 0/0 drop: 0/0

Overhead	Shared Object	Symbol
9.87%	[kernel]	[k] entry_SYSEXIT_unsafe_stack
8.32%	[kernel]	[k] __pte_offset_map
7.45%	[kernel]	[k] switch_mm_irqs_off
7.17%	[kernel]	[k] sched_clock_noinstr
6.90%	[kernel]	[k] entry_SYSCALL_64
6.81%	perf	[.] 0x000000000000a9632
6.80%	[kernel]	[k] psi_group_change
6.73%	[kernel]	[k] task_contending
6.71%	[kernel]	[k] native_write_msr
6.60%	[kernel]	[k] dl_server_update_idle_time
6.29%	[kernel]	[k] update_sd_lb_stats.constprop.0
4.49%	[kernel]	[k] update_load_avg
2.46%	[kernel]	[k] dequeue_task_fair
2.01%	[kernel]	[k] ktime_get_update_offsets_now
1.91%	libc.so.6	[.] handle_fildes_io
1.69%	[kernel]	[k] hrtimer_get_next_event
1.61%	[kernel]	[k] printk_get_next_message
1.26%	[kernel]	[k] __switch_to
1.00%	[kernel]	[k] place_entity
0.96%	[kernel]	[k] vsnprintf
0.77%	[kernel]	[k] dl_scaled_delta_exec
0.53%	[kernel]	[k] smp_call_function_single
0.49%	[kernel]	[k] x64_sys_call
0.30%	[kernel]	[k] perf_event_for_each_child
0.23%	[kernel]	[k] event_function
0.18%	[kernel]	[k] sched_clock
0.17%	[kernel]	[k] end_repeat_nmi
0.11%	[kernel]	[k] nmi_restore
0.05%	[kernel]	[k] perf_sample_event_took
0.05%	[kernel]	[k] native_sched_clock
0.03%	[kernel]	[k] nmi_cpu_backtrace
0.02%	[kernel]	[k] perf_event_nmi_handler
0.02%	[kernel]	[k] intel_bts_enable_local
0.01%	[kernel]	[k] intel_pmu_handle_irq

For a higher level overview, try: perf top --sort comm,dso